Paper Chromatography

Exam code: 0620 & 0971

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Updated on 24 September 2024

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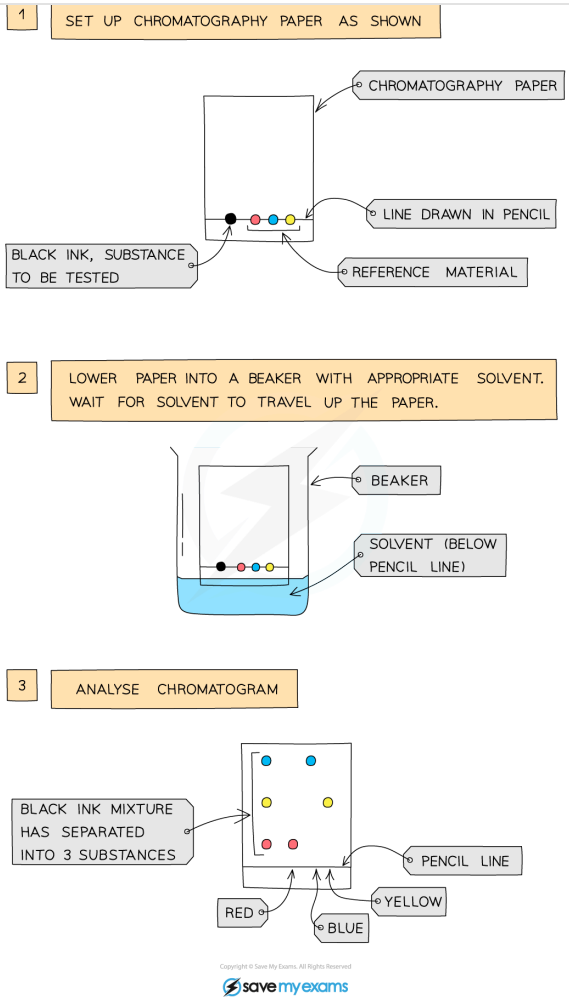
Paper chromatography

- Chromatography is used to separate substances and provide information to help identify them
- The components have **different solubilities** in a given solvent
  - E.g. Different coloured inks that have been mixed to make black ink
- A **pencil line** is drawn on chromatography paper and spots of the sample are placed on it
  - A pencil is used for this as ink would run into the chromatogram along with the samples
- The paper is then lowered into the solvent container, making sure that the pencil line sits **above** the level of the solvent so the samples don't wash into the solvent container
  - The solvent used is usually **water** but it can be other substances such as **ethanol**
- The solvent travels up the paper by **capillary action**, taking some of the coloured substances with it
- Different substances have different solubilities so they will travel at different rates, causing the substances to spread apart
  - Those substances with **higher** solubility will travel further than the others

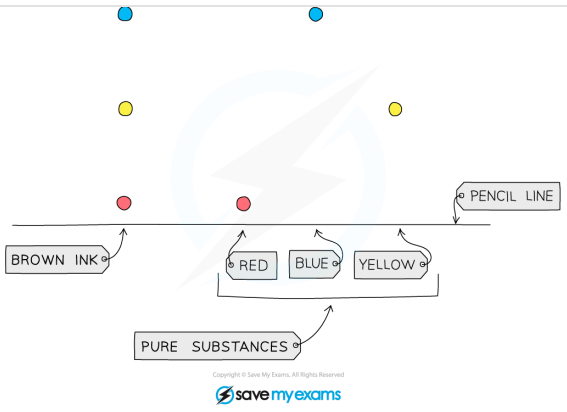
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



The pigments in ink can be analysed using paper chromatography

Interpret simple chromatograms

- We can use a chromatogram to compare the substances present in a mixture to known substances and make assumptions
  - Pure substances will produce only **one spot** on the chromatogram
  - Impure substances will produce **more than one spot** on the chromatogram
  - If two or more substances are the same, they will produce identical chromatograms
  - If the substance is a **mixture**, it will separate on the paper to show all the **different components** as **separate** spots
- It is common practice to include a known compound as a reference spot
  - This can help match up to an unknown spot or set of spots in order to identify it



The brown ink has separated showing a spot of red ink, blue ink and yellow ink

- We can draw several **conclusions** from this chromatogram:
  - The brown ink is a mixture as there are three dots
  - Red, yellow and blue are pure as there is only one dot for each
  - The brown ink contains red, blue and yellow as the dots are in line with one another horizontally

Examiner Tips and Tricks

Chromatograms in exams will be in black and white so to identify whether a mixture contains a known sample, the dots need to be in line with one another.

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